

ZAHIR JACOB

ARTIFICIAL INTELLIGENCE STUDENT · MACHINE LEARNING, COMPUTER VISION & NLP

PROFILE

Final-semester student in the University Technical Degree in Artificial Intelligence (FCEIA-UNR). I build machine learning and NLP systems end to end: data exploration, model selection and validation, reproducible inference and deployment. Solid foundation in databases, ETL and business intelligence.

SKILLS

Programming
Python (Pandas, NumPy, Matplotlib), SQL, JavaScript, HTML, CSS

Machine Learning
Scikit-learn, TensorFlow/Keras, PyTorch, cross-validation, temporal validation, hyperparameter tuning, metrics (F1, ROC-AUC)

NLP & LLMs
RAG, hybrid retrieval (embeddings + BM25), re-ranking, LangChain, Hugging Face, ChromaDB, ReAct agents, Gradio

Computer Vision
OpenCV, segmentation, contours, OCR

Data & BI
SQL Server, PostgreSQL, SSIS, SSMS, ETL, web scraping, data warehousing, dimensional modeling, Power BI

Tools & Methods
Git/GitHub, GitHub Actions, Docker, Streamlit, Jupyter, Render · Scrum, Kanban

LANGUAGES

Spanish: native
English: upper-intermediate (advanced comprehension, functional speaking)

CONTACT

Rosario, Argentina
Relocating to Copenhagen in 2027
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EDUCATION

University Technical Degree in Artificial Intelligence
FCEIA – National University of Rosario (UNR) | 2023 – present

- Final semester.
- Areas: Machine Learning, Data Mining, Databases, Business Intelligence, Computer Vision and NLP.

SELECTED PROJECTS

Multi-source RAG assistant with a ReAct agent
Python · LangChain · ChromaDB · BM25 · Local LLM | github.com/ZahirJacob/multi-source-rag-assistant

- Conversational assistant over documents, tabular data and a knowledge graph, with hybrid retrieval (semantic + BM25 + cross-encoder re-ranking).
- The LLM writes JSON filters and Cypher queries; conversational memory and a ReAct agent with tools. Runs entirely on local models.
- Intent classifier reaching 0.96 macro F1 on the test set.

Price-intelligence platform for a phone-repair shop (real client)
Python · PostgreSQL · GitHub Actions · Streamlit | github.com/ZahirJacob/repuestos-radar

- Daily automated ingestion from 8 vetted reseller storefronts into a Postgres price history, with polite scraping (robots.txt, honest user-agent, backoff).
- Dashboard with current prices, price history and repair margins; the client manages the watchlist without touching code. Tested with CI.
- Also built and delivered the shop's website (activcelu.com.ar).

Australian rain prediction + deployed web app
Python · Scikit-learn · TensorFlow · Docker · Render | australia-rain-app.onrender.com

- Classifier trained on 10 years of observations (142,153 rows, 49 stations): 23 configurations compared with stratified cross-validation and expanding-window temporal evaluation.
- Selected model: F1 0.640 (temporal validation), ROC-AUC 0.885 (test). Deployed as a live app that fetches today's weather and runs the model in pure NumPy; Docker inference.

Automatic license-plate recognition
Python · OpenCV · OCR

- Localization, segmentation and character-recognition pipeline (Otsu, black-hat, Sobel); all seven characters correct in 8 of 12 images.

Data warehouse & sales dashboard
SQL Server · SSIS · SSMS · Power BI

- Multi-source ETL into a data warehouse (facts, dimensions, stored procedures) and a Power BI dashboard with sales and Pareto analysis.